



Title: WHAT IS ENCRYPTION? HOW DOES IT WORKS?

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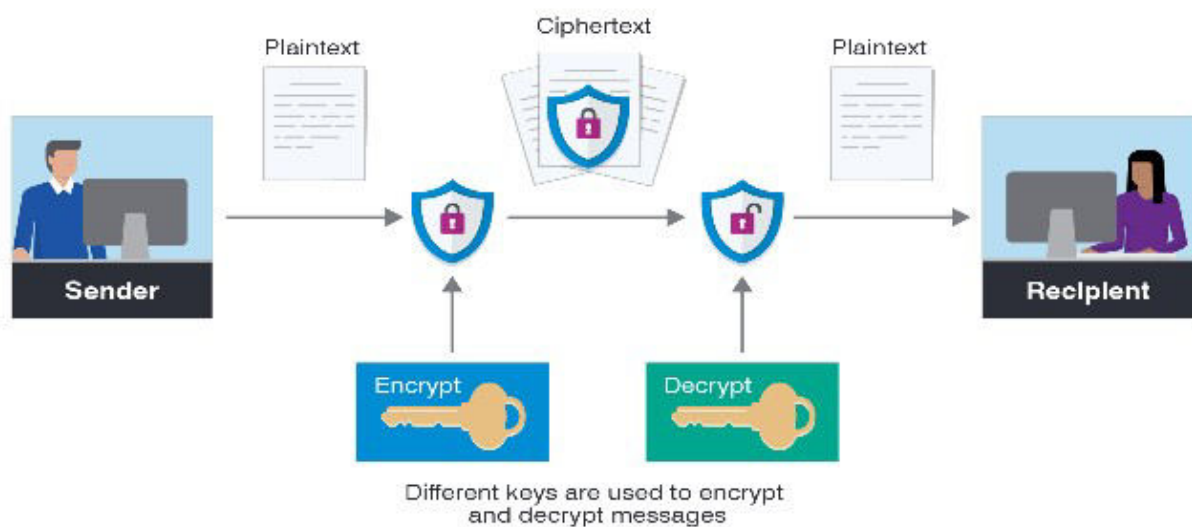
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What is Encryption?

Encryption is a way of translating data from plaintext (unencrypted) to ciphertext (encrypted). Users can access encrypted data with an encryption key and decrypted data with a decryption key. Protecting your data.



How does Encryption work?

Encryption takes plain text, like a text message or email, and scrambles it into an unreadable format called “cipher text.” This helps protect the confidentiality of digital data either stored on computer systems or transmitted through a network like the Internet.

When the intended recipient accesses the message, the information is translated back to its original form. This is called decryption.

To unlock the message, both the sender and the recipient have to use a “secret” encryption key a collection of algorithms that scramble and unscramble data back to a readable format.

What are the two main types of encryption systems?

An encryption key is a series of numbers used to encrypt and decrypt data. Encryption keys are created with algorithms. Each key is random and unique.

There are two main types of encryption systems:

symmetric encryption and
asymmetric encryption. Here's how they're different.

Symmetric encryption uses a single password to encrypt and decrypt data.

Asymmetric encryption uses two keys for encryption and decryption. A public key, which is shared among users, encrypts the data. A private key, which is not shared, decrypts the data.

What is an encryption algorithm?

An encryption algorithm is the set of rules, usually governing a computer or other tech device such as a smart phone, that turns readable data into scrambled cipher text.

The data scrambled by these algorithms look like randomized code. But the algorithms configure this scrambled data in a purposeful way so that it can easily be turned back into a readable format by a decryption key.

What are the types of encryption algorithms?

There are several types of encryption, Here are the most common examples of encryption.

AES is a symmetric block cipher chosen by the U.S. government to protect classified information; it is implemented in software and hardware throughout the world to encrypt sensitive data. NIST started development of AES in 1997 when it announced the need for a successor algorithm for the Data Encryption Standard ([DES](#)), which was starting to become vulnerable to [brute-force](#) attacks.

- **DES** is an outdated symmetric key method of data encryption. DES works by using the same key to encrypt and decrypt a message, so both the sender and the receiver must know and use the same private key. DES has been superseded by the more secure AES algorithm.
- [Diffie-Hellman key exchange](#), also called exponential key exchange, is a method of digital encryption that uses numbers raised to specific powers to produce decryption keys on the basis of components that are never directly transmitted, making the task of a would-be code breaker mathematically overwhelming.
- **Elliptical curve cryptography ([ECC](#))** uses algebraic functions to generate security between key pairs. The resulting cryptographic algorithms can be faster and more efficient and can produce comparable levels of security with shorter cryptographic keys. This makes ECC algorithms a good choice for internet of things ([IoT](#)) devices and other products with limited computing resources.
- **Quantum key distribution (QKD)** is a proposed method for encrypted messaging by which encryption keys are generated using a pair of entangled photons that are then transmitted separately to the message. Quantum entanglement enables the sender and receiver to know whether the encryption key has been intercepted or changed before the transmission even arrives. This is because, in the quantum realm, the very act of observing the transmitted information changes it. Once it has been determined that the encryption is secure and has not been intercepted, permission is given to transmit the encrypted message over a public internet channel.
- **RSA** was first publicly described in 1977 by Ron Rivest, Adi Shamir and Leonard Adleman of the Massachusetts Institute of Technology (MIT), though the 1973 creation of a public key algorithm by British mathematician Clifford Cocks was

kept classified by the U.K.'s Government Communications Headquarters (GCHQ) until 1997. Many protocols, like Secure Shell ([SSH](#)), OpenPGP, Secure/Multipurpose Internet Mail Extensions (S/MIME) and Secure Sockets Layer ([SSL](#))/TLS, rely on RSA for encryption and [digital signature](#) functions.